MOBILE DEVELOPMENT MINI-PROJECT 1

VKU Field Survey

Offline Data Collection PWA and Capacitor

> A local-first campus facility inspection system for classrooms, laboratories, basements and remote buildings with weak or unavailable connectivity.

1. Problem And Objective

A network-dependent inspection form can lose unfinished observations, photos and submissions. This project treats IndexedDB as the primary source of truth: it autosaves drafts, accepts submissions offline, survives refresh/restart and synchronizes queued UUID records when the API is reachable.

The professional extension adds inspector/session accountability, category checklists, issue classification, versioned edits, GPS evidence, Admin review and Google Sheets reporting while preserving the original PWA and sync requirements. A server-validated Admin access key gates the review screen in both the PWA and APK.

2. Feature Checklist

DONE	Standalone manifest, VKU theme, 192/512 icons and app-shell precache
DONE	IndexedDB autosave, draft restore and schema-v1 compatibility
DONE	Local profile, active session and immutable survey snapshots
DONE	Six-step form, custom room, room type and 1-5 star rating
DONE	Five category-specific checklists with OK/Issue/Not Checked
DONE	Severity, priority, issue type, recommended action and evidence rules
DONE	Photo/GPS evidence with web and Capacitor implementations
DONE	Sequential offline queue, reconnect sync and idempotent UUID API
DONE	Filters, Needs Action badge, dashboard and local CSV/JSON export
DONE	Stable-UUID edits, version history and refresh-safe edit drafts
DONE	Admin assignment, review status, notes and central export
DONE	Server-validated Admin access key gate for PWA and APK
DONE	Google Sheets/Drive upsert, evidence URL and AuditLog
DONE	Java 21 APK build and public GitHub Release download
DONE	Public HTTPS PWA and same-origin Vercel API deployment
PENDING	Physical-device Camera, Network and GPS verification

SYSTEM DESIGN

Architecture And Offline Flow

3. Architecture

```
React UI (shared PWA / Android code)
  -> IndexedDB v3: surveys | profile | sessions | surveyEdits
  -> local-first submit: DRAFT -> PENDING_SYNC
  -> foreground sync / Background Sync (sequential)
  -> Express / Vercel REST API
  -> Apps Script -> Google Sheets + Drive -> SYNCED
```

Client

React, TypeScript, Vite, Tailwind, React Router, `idb`, Workbox and Capacitor.

Server

Node.js, Express/Vercel API, Admin key middleware, version validation and an Apps Script central bridge.

4. Data Model

Each Survey includes UUID/version, sync and review statuses, inspector snapshot, session snapshot, facility location, room type, category checklist, rating, severity, priority, issue type, recommended action, notes, optional Base64 photo and GPS status/evidence.

Snapshots preserve historical accountability when the local profile or active session changes. IndexedDB version 3 also stores autosaved edits; a runtime normalizer safely reads original survey records.

5. Reliability Rules

- Draft changes are debounced and stored locally.
- Submission persists locally before any API request.
- Only a confirmed response changes status to Synced.
- Failures retain all data and remain retryable.
- A lock prevents overlapping foreground sync runs.
- Queued records upload in creation order; repeated UUIDs do not duplicate server data.
- Submitted edits retain UUID, increment version and return to Pending Sync.
- Admin review metadata stays separate from device synchronization state.
- Startup, browser/native network events and manual retry back up Background Sync.

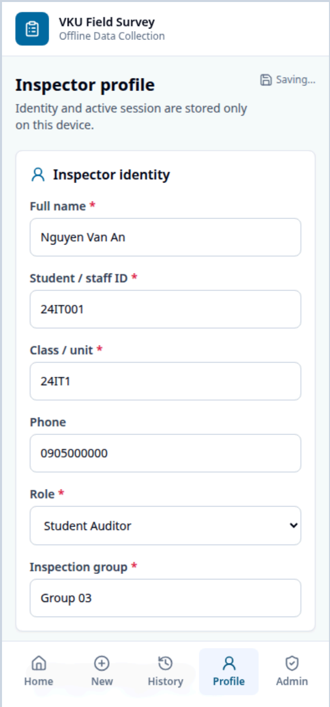
6. PWA And Native Bridge

Workbox precaches HTML, CSS, JavaScript, icons and manifest data for offline boot. Browser evidence uses file input and Web Geolocation; Android uses Capacitor Camera, Network and Geolocation. GPS failure is recorded as unavailable and does not block a valid survey.

Java 21/Gradle builds package `edu.vku.fieldsurvey` for Android SDK 23-35. GitHub Actions signs and publishes the debug APK, so Android Studio is not required on the development machine.

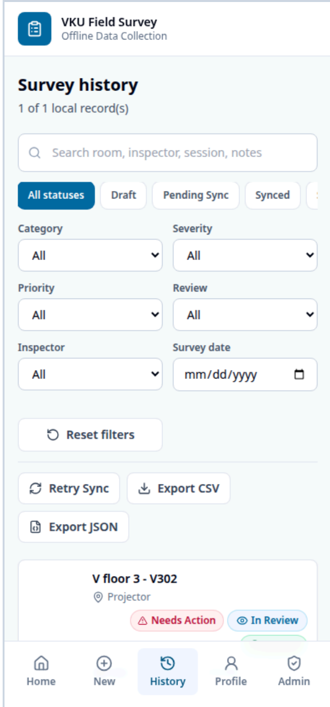
IMPLEMENTED INTERFACE

Key Screens

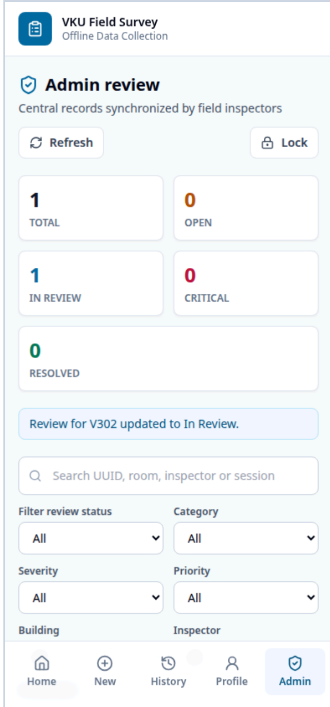


Offline inspector identity

Six-step inspection review and validation



Professional filters, sync and local export

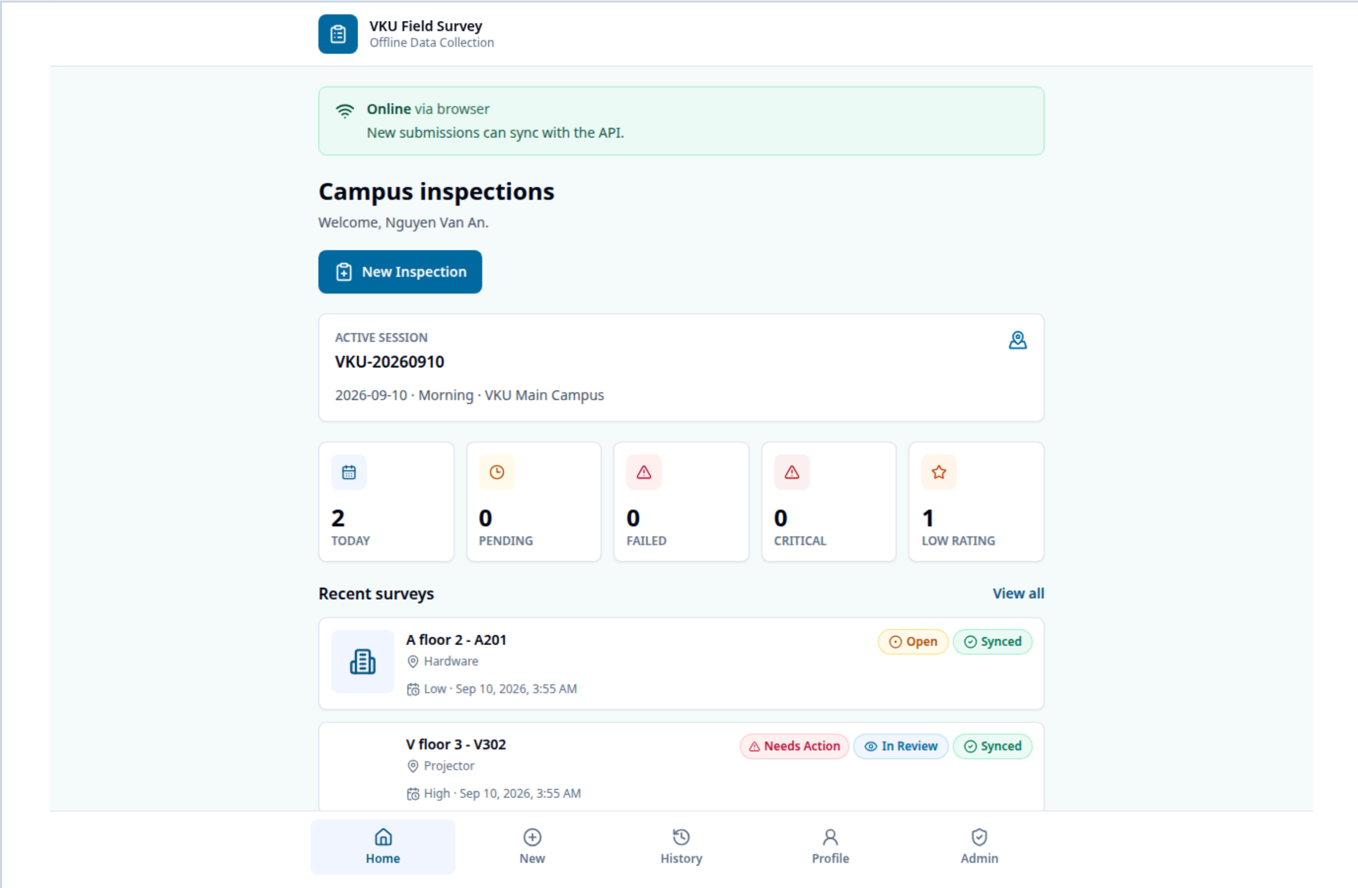


Central Admin review and assignment

Screenshots are generated by the repeatable Chromium E2E workflow at a 390 x 844 mobile viewport.

VERIFICATION AND DELIVERY

Results



7. Automated Verification

Unit / Integration

Business rules, High/Critical photo requirement, old-schema normalization, IndexedDB edit recovery, immutable snapshots, queue ordering and server/review validation.

Browser E2E

Profile refresh, edit versioning, Admin review, online/offline submit, reconnect sync, GPS, export, wrong-key rejection and offline boot.

8. Delivery Status

The PWA and same-origin API are live at `miniproject1-client.vercel.app`. Source is public at `github.com/tuaw-khoi/miniproject1`; the `apk-latest` Release provides the signed Android APK. Production verification covers deep links, CORS, versioned idempotent sync, Google Sheets central records, service-worker offline reload and APK metadata. Physical-device plugin testing remains. The demo Admin key is `VKU-ADMIN-2026` and can be replaced through `ADMIN_ACCESS_KEY`.

9. Conclusion

VKU Field Survey meets the original offline-first learning objectives and now models a credible inspection-to-resolution process. Its core guarantee remains unchanged: valid field data is saved locally first and never depends on immediate connectivity.